

Adaptive norm-based encoding of within-person facial variability



Yaren Koca & Chris Oriet
Dept. of Psychology, University of Regina



What is the architecture of familiar face representations?

According to the face space theory (Valentine, 1991), faces are encoded in a multidimensional space based on their perceived characteristics like age, eye shape, expressions, etc.

Each person’s facial identity varies idiosyncratically across such characteristics (Jenkins et al., 2011), which is incompatible with the face space’s assumption that all faces are encoded across common dimensions.

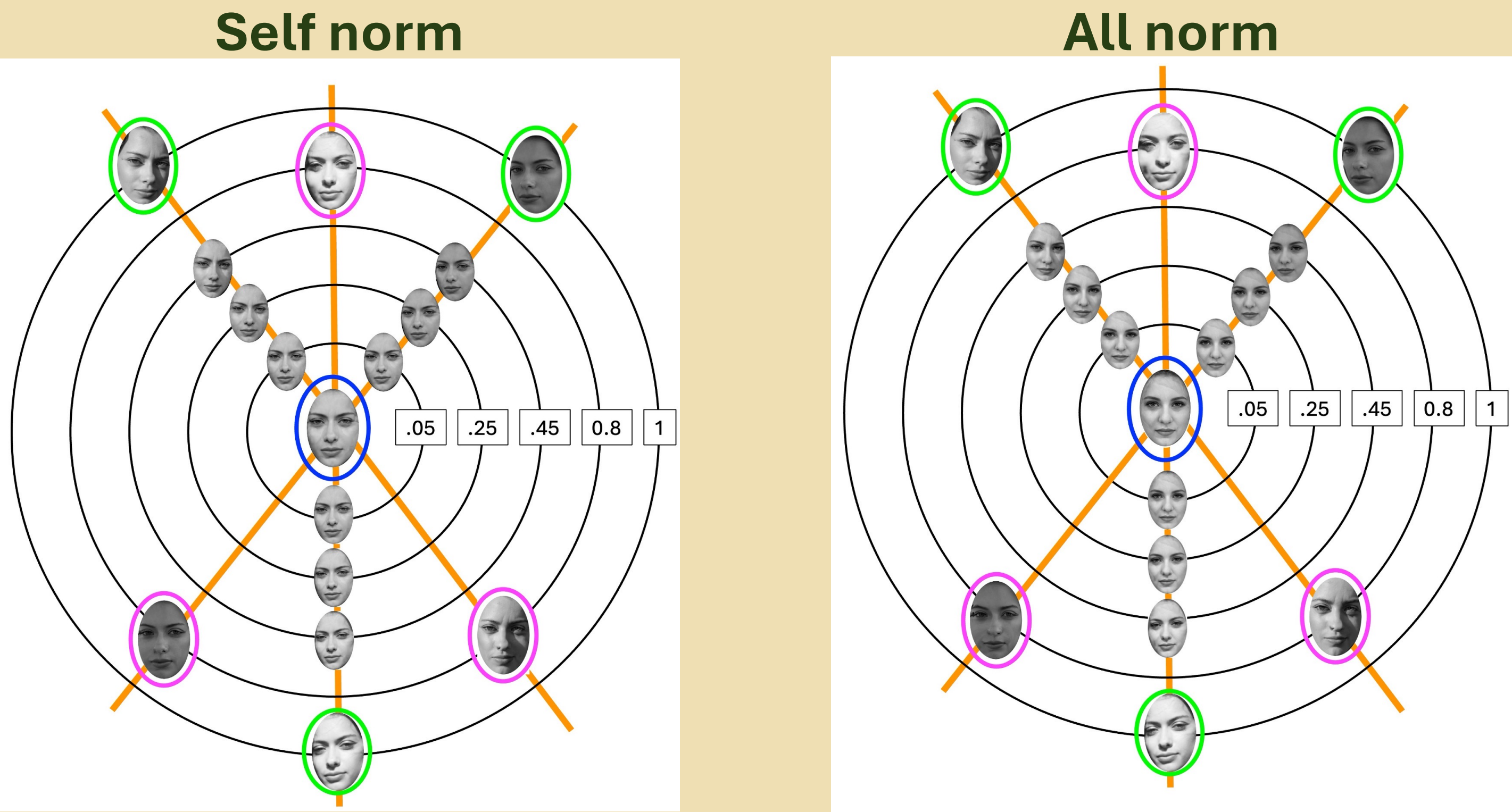
How do we represent the idiosyncratic variability of familiar faces in the face space?

One possibility is that different views of faces are encoded as vectors in reference to an identity-specific average (Koca & Oriet, 2023) in private face spaces (Laurence et al., 2022).

This is akin to traditional norm-based models of the face space that suggest different faces are encoded in reference to a norm face.

Variability across different views of familiar faces are encoded in reference to **identity-specific averages**.

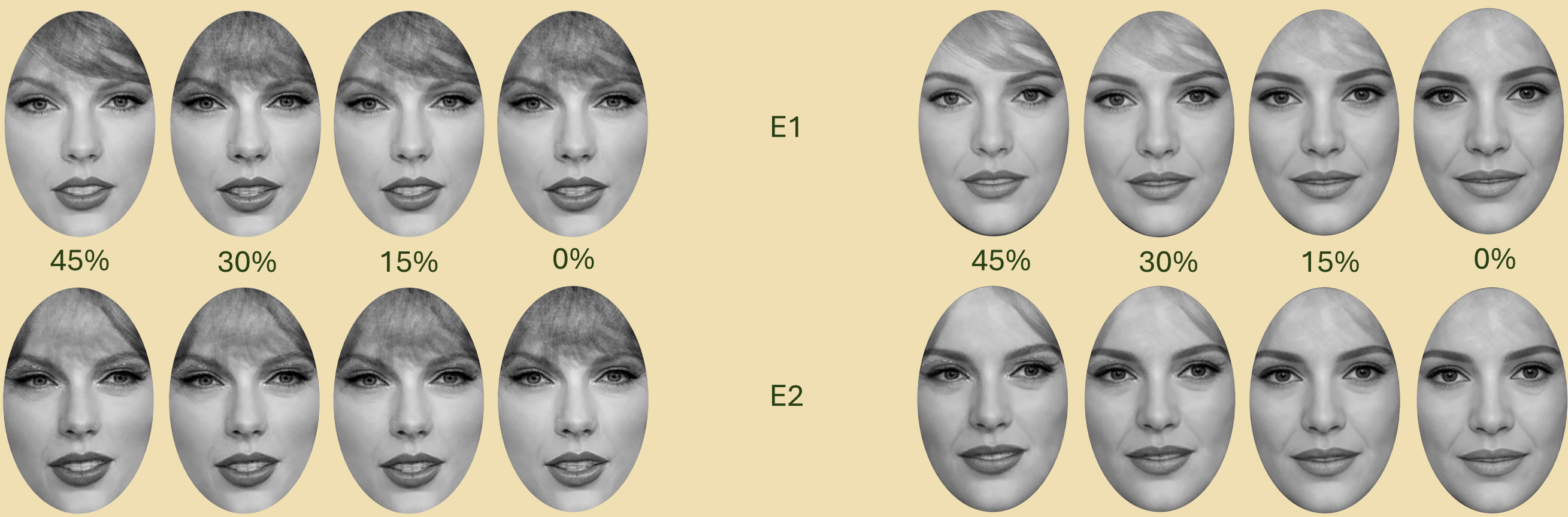
Methods



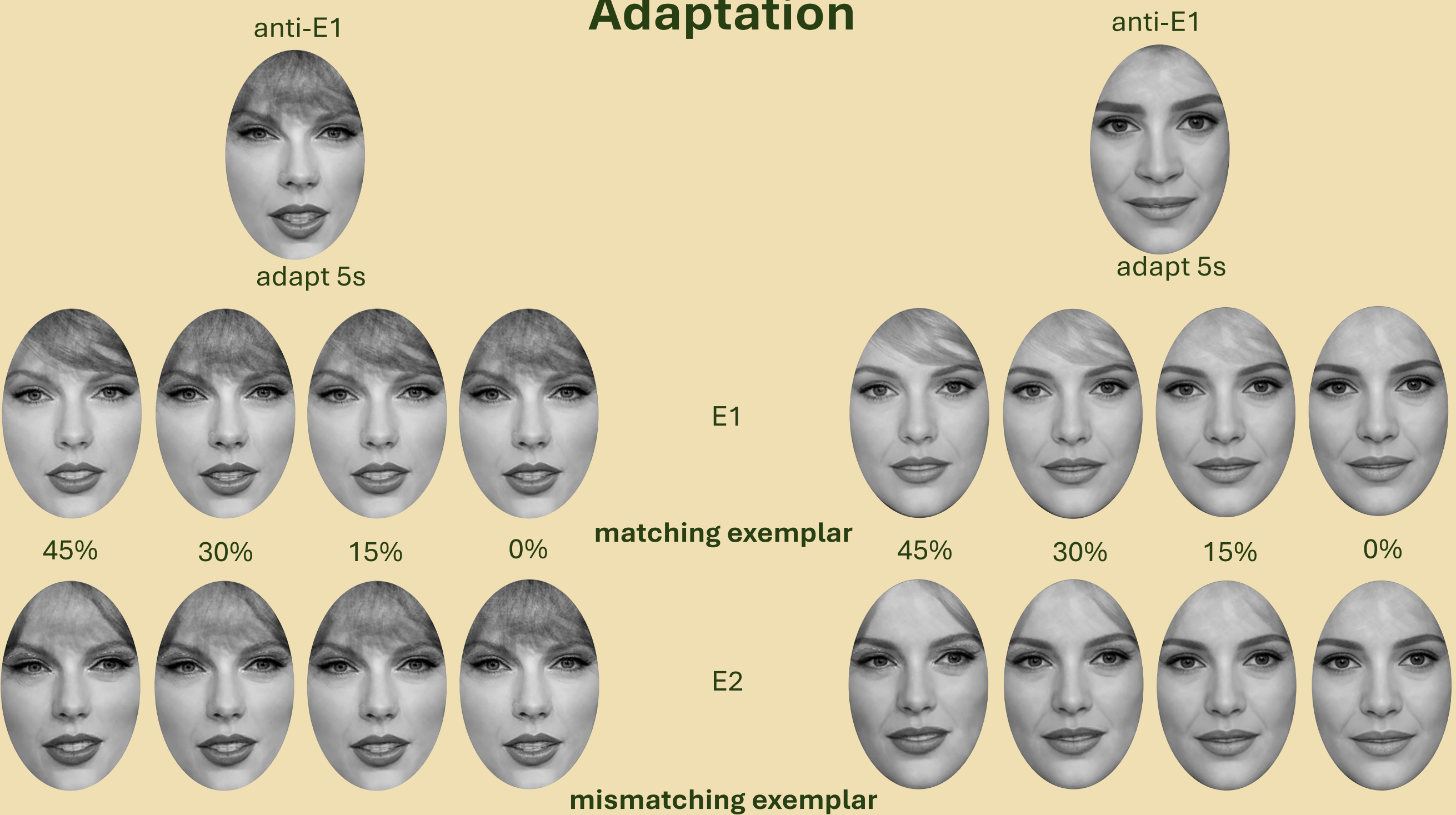
Practice



Baseline

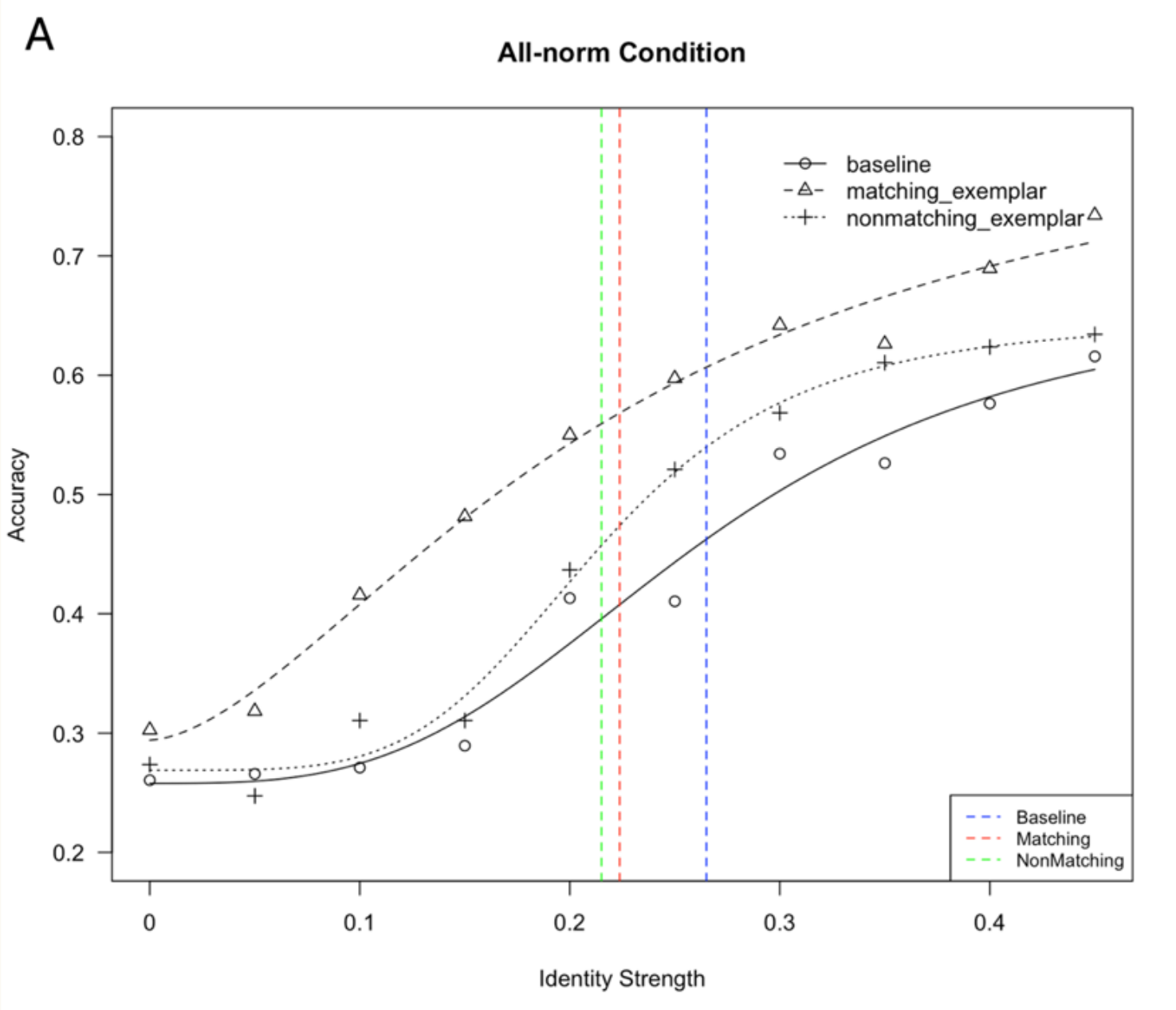
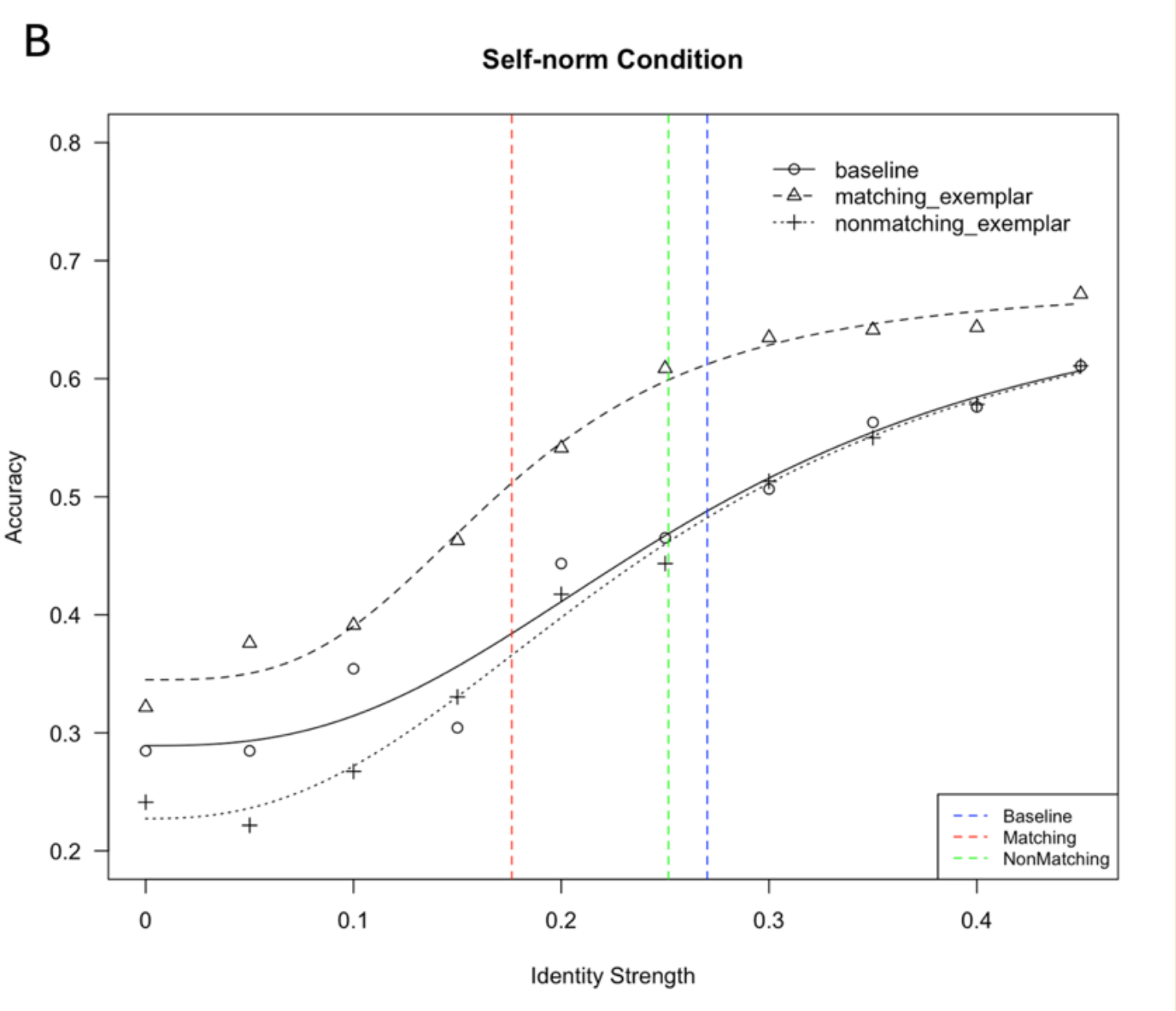


Adaptation



Results

Independent variables	
Anti-face (between-subjects)	Self norm, All norm
Adaptation (within-subjects)	Baseline, Matching Exemplar, Mismatching Exemplar
Image Strength (within-subjects)	0%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%
Dependent variable: Identification threshold	



9.4%

4.1%

N = 42

Adaptation effects were larger for matching exemplars in the self-norm condition than the all-norm condition.

This perceptual shift in the self-norm condition was systematic; identification of mismatching exemplars did not benefit from adaptation to the same extent.

Adaptation to an anti-image shifted the perception of the average images in the direction of the matching exemplar.

Discussion

Our data support the averaging hypothesis while explaining why we average facial identity. Facial identity may serve as a reference for which within-person variability is encoded for familiar faces.

Exemplar-based models would predict larger adaptation effects as a function of similarity between adaptor-test, which is not supported by the current data (smaller adaptation effect for mismatching exemplars in the self-norm condition).

Facial identity may be a category in the face space akin to age, gender, ethnicity (Hole, 2011; Laurence et al., 2022).

Limitations:

Differentiation of images of the same person is limited in ecological validity.

Limited number of identities limits generalizability (Burton, 2025).

Unclear whether the results would replicate with unfamiliar faces.

Task: Identify different images of the same familiar identity